

Analysis 2
Serie 12
Lukas Mengis

1.

We have:

$$(\gamma \star \sigma)(t) := \begin{cases} \sigma(2t) & \text{for } 0 \leq t \leq \frac{1}{2} \\ \gamma(2t-1) & \text{for } \frac{1}{2} \leq t \leq 1 \end{cases}$$

Using the definition of line integrals

$$\begin{aligned} \int_{\gamma \star \sigma} F \cdot d(\gamma \star \sigma) &= \int_0^1 F(\gamma \star \sigma(t)) \cdot (\gamma \star \sigma)'(t) dt \\ &= \int_0^{\frac{1}{2}} F(\sigma(2t)) 2\sigma'(t) dt + \int_{\frac{1}{2}}^1 F(\gamma(2t-1)) 2\gamma'(t) dt \\ &= \int_0^1 F(\sigma(u) + \sigma'(u)) du + \int_0^1 F(\gamma(z) + \gamma'(z)) dz \\ &= \int_{\sigma} F \cdot d\sigma + \int_{\gamma} F \cdot d\gamma \end{aligned}$$

2.

2.a.

Define a homotopy H

$$H : [0, 1] \times [0, 1] \rightarrow U$$

Define the points x and p

$$H(0, s) = x \quad H(1, s) = p$$

Then the every path can be written using the homotopy

$$\gamma_s(t) = H(t, s)$$

$$\Rightarrow f(s) = \int_{\gamma_s} F \cdot d\gamma_s = \int_0^1 F(H(t, s)) \cdot H_t(t, s) dt$$

Where $H_i = \frac{\partial H}{\partial i}$

We want to show that $f'(s) = 0$. This is the case if f is well-posed.

$$\begin{aligned} f'(s) &= \frac{d}{ds} \int_0^1 F(H(t, s)) \cdot H_t(t, s) dt \\ &= \int_0^1 \frac{\partial}{\partial s} (F(H(t, s)) \cdot H_t(t, s)) dt \\ &= \int_0^1 \sum_{i=1}^n \frac{\partial}{\partial s} (F_i(H) \cdot (H_t)_i) dt \end{aligned}$$

Product rule

$$\frac{\partial}{\partial s} (F_i(H) \cdot (H_t)_i) = \frac{\partial}{\partial s} F_i(H) (H_t)_i + F_i(H) (H_t)_{ts}$$

And chain rule

$$\frac{\partial}{\partial s} (F_i(H)) = \sum_{j=1}^n \frac{\partial F_i}{\partial x_j} (H) (H_s)_j$$

Since $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ the order of j, i does not matter

$\Rightarrow$

$$f'(s) = \int_0^1 \left(\sum_{0 < j, i \leq n} \frac{\partial F_i}{\partial x_j}(H)(H_t)_i(H_s)_j + \sum_{0 < i \leq n} F_i(H)(H_i)_{ts} \right) dt$$

This means

$$\frac{\partial}{\partial s} F(H(t, s)) \cdot H_t(t, s) = \sum_{0 < j, i \leq n} \frac{\partial F_i}{\partial x_j}(H)(H_t)_i(H_s)_j = \frac{\partial}{\partial t} F(H(t, s)) \cdot H_s(t, s)$$

$$\Rightarrow f'(s) = [F(H(t, s)) \cdot H_s(t, s)]_0^1$$

$$H(0, s) = p \quad H(1, s) = x$$

$$H_s(0, s) = H_s(1, s) = 0$$

The points are fixed

$$\Rightarrow f'(s) = 0$$

2.b.

$$f(x) = \int_{\gamma_x} F \cdot d\gamma_x$$

$$f(x') = \int_{\gamma_{x'}} F \cdot \gamma_{x'} = \int_{\gamma_x \star \sigma} F \cdot d(\gamma_x \star \sigma) = \int_{\gamma_x} F \cdot d\gamma_x + \int_{\sigma} F \cdot d\sigma$$

$$\Rightarrow f(x) - f(x') = - \int_{\sigma} F \cdot d\sigma$$

2.c.

3.

3.a.

It's connected but not simply connected. The z-axis is removed from the domain.

3.b.

$$\begin{aligned} \text{rot}(F) &= \nabla \times F = \begin{pmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{pmatrix} \times F(x, y, z) \\ &= \begin{pmatrix} \frac{1}{x^2+y^2} 2y - \frac{2y}{x^2+y^2} \\ \frac{2x}{x^2+y^2} - \frac{2x}{x^2+y^2} \\ \frac{-2(x^2+y^2)-2x(2(yz-x))}{(x^2+y^2)^2} - \frac{2(x^2+y^2)-2y(2(xz+y))}{(x^2+y^2)^2} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

$\Rightarrow F$ is conservative

$$\begin{aligned}
\int_{\gamma_1} F \cdot d\gamma_1 &= \int_0^{2\pi} F(\cos(t), \sin(t), 0) \cdot (-\sin(t), \cos(t), 0) dt \\
&= \int_0^{2\pi} (2\sin(t), -2\cos(t), 0) \cdot (-\sin(t), \cos(t), 0) dt \\
&= \int_0^{2\pi} -2 dt = -4\pi
\end{aligned}$$

3.c.

From stokes $\Rightarrow$ for every oriented area S

$$\int_{\partial S} F \cdot d\gamma = \int_S \text{rot}(F) \cdot dS = 0$$

Since we can get an area which border is equal to the two path γ_1, γ_2 and does not intersect the z -axis:

$$\begin{aligned}
0 &= \int_{\partial S} F d\gamma = \int_{\gamma_1} F \cdot d\gamma_1 + \int_{\overline{\gamma_2}} F \cdot d\overline{\gamma_2} \\
&\Rightarrow \int_{\gamma_1} F \cdot d\gamma_1 = \int_{\gamma_2} F \cdot d\gamma_2
\end{aligned}$$

3.d.

no

4.

4.a. false

4.b. true

4.c. false

4.d. true

5.

$$F(x, y) = \begin{pmatrix} \lambda x e^y \\ (y + 1 + x^2) e^y \end{pmatrix}$$

Put the function in $\mathbb{R}^3$. Now we can simply calculate $\text{rot}(F')$

$$\text{rot}(F') = \Delta \times F' = \begin{pmatrix} \frac{d}{dx} \\ \frac{d}{dy} \\ \frac{d}{dz} \end{pmatrix} \times \begin{pmatrix} \lambda x e^y \\ (y + 1 + x^2) e^y \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2x e^y - \lambda x e^y \end{pmatrix}$$

The function F is conservative if $\text{rot}(F) = 0$

$$\Rightarrow 2x e^y = \lambda x e^y \Rightarrow \lambda = 2$$

We want to find a potential φ

$$\Delta \varphi = F$$

$$\varphi(x, y) = \int 2x e^y dx = x^2 e^y + C(y)$$

$$\begin{aligned}
\frac{\partial \varphi}{\partial y} &= x^2 e^y + C'(y) = (y + 1 + x^2) e^y \\
&\Rightarrow C'(y) = (y + 1) e^y \\
&\Rightarrow C(y) = y e^y + c \\
&\Rightarrow \varphi(x, y) = (y + x^2) e^y + c
\end{aligned}$$

6.

$$\begin{aligned}
\int_S \text{rot}(F) \cdot dn &= \int_S \begin{pmatrix} -2z \\ 0 \\ -x \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} dS \\
&= \int_S -3xz \cdot dS \\
&= -3 \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin(\varphi) \cos(\theta) \cos(\varphi) \sin(\varphi) d\varphi d\theta \\
&= 0 \\
\int_S \text{rot}(F) n dS &= \int_{\partial S} F \cdot d\gamma
\end{aligned}$$

with

$$\begin{aligned}
\gamma(t) &= (\cos(t), \sin(t), 0) \\
\gamma'(t) &= (-\sin(t), \cos(t), 0) \\
F(\gamma(t)) &= (\sin(t) \cos(t), 0, 1)
\end{aligned}$$

$$\int_0^{2\pi} (\sin(t) \cos(t), 0, 1) \cdot (-\sin(t), \cos(t), 0) dt = \int_0^{2\pi} -\sin^2(t) \cos(t) dt$$

cos and sin are 2π periodically

$$\int_S \text{rot}(F) \cdot dn = 0$$

7.

7.a.

$$\text{rot}(\varphi f) = \nabla \times (\varphi f) = (\nabla \varphi) \times f + \varphi \nabla f = (\nabla \varphi) \times f + \varphi \text{rot}(f)$$

7.b.

$$\text{div}(\varphi f) = \sum_{i=1}^3 \frac{\partial}{\partial x_i} (\varphi_i f_i) = \sum_{i=1}^3 \frac{\partial \varphi}{\partial x_i} f_i + \varphi \frac{\partial f_i}{\partial x_i} = \langle \nabla \varphi, f \rangle + \varphi \text{div}(f)$$

7.c.

$$\text{div}(\text{rot}(f)) = \partial_x (\partial_y f_z - \partial_z f_y) + \partial_y (\partial_z f_x - \partial_x f_z) + \partial_z (\partial_x f_y - \partial_y f_x) = 0$$

7.d.

$$\text{rot}(\nabla\varphi) = \begin{pmatrix} \partial_y\partial_z\varphi - \partial_z\partial_y\varphi \\ \partial_z\partial_x\varphi - \partial_x\partial_z\varphi \\ \partial_x\partial_y\varphi - \partial_y\partial_x\varphi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

8.

$$(a) = -(b)$$

9.

9.a.

let

$$\omega = d\psi^1 \wedge \dots \wedge d\psi^n$$

Each $d\psi^i$ is a function of $(x_1, \dots, x_n)$

$$d\psi^j = \sum_{i=1}^n \frac{\partial\psi^j}{\partial x_i} dx^i$$

$$\omega = \sum_{i_1=1}^n \frac{\partial\psi^1}{\partial x_{i_1}} dx^{i_1} \wedge \dots \wedge \sum_{i_n=1}^n \frac{\partial\psi^n}{\partial x_{i_n}} dx^{i_n}$$

Since $dx^i \wedge dx^i = 0$ only permutations using all $dx, dx^2, \dots, dx^n$ survive.

Denote σ as such a permutation.

$$\sum_{\sigma \in S_n} \left(\prod_{i=1}^n \frac{\partial\psi^i}{\partial x_{\sigma(i)}} \right) dx^{\sigma(1)} \wedge \dots \wedge dx^{\sigma(n)}$$

The order matters

$$\begin{aligned} \omega &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) \left(\prod_{i=1}^n \frac{\partial\psi^i}{\partial x_{\sigma(i)}} \right) dx^1 \wedge \dots \wedge dx^n \\ &= \det \left(\frac{\partial\psi_j}{\partial x_i} \right) dx^1 \wedge \dots \wedge dx^n \end{aligned}$$

9.b.

$$\begin{aligned} \Phi : U \subset \mathbb{R}^n &\rightarrow \mathbb{R}^n \\ x &\mapsto \Phi(x) = y \end{aligned}$$

$$\int_{\psi(U)} f(y) dy^1 \wedge \dots \wedge dy^n = \int_U f(\psi(x)) \det(D\psi(x)) dx^1 \wedge \dots \wedge dx^n = \int_U f(\psi(x)) d\psi^1 \wedge \dots \wedge d\psi^n$$